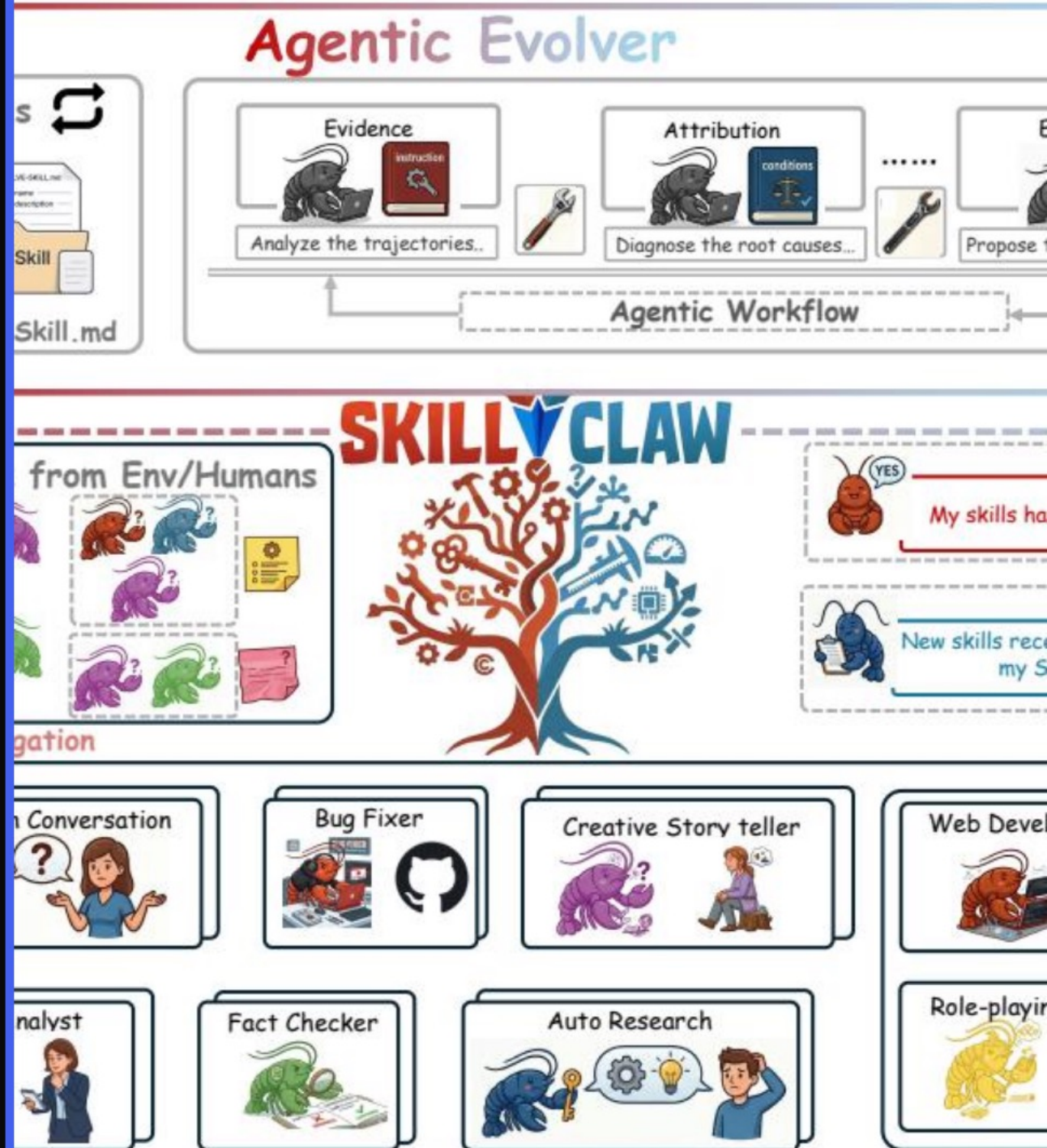


SkillClaw: Let Skills Evolve Collectively with Agentic Evolver

arxiv.org/abs/2604.08377



Agent Skills Remain Static After Deployment

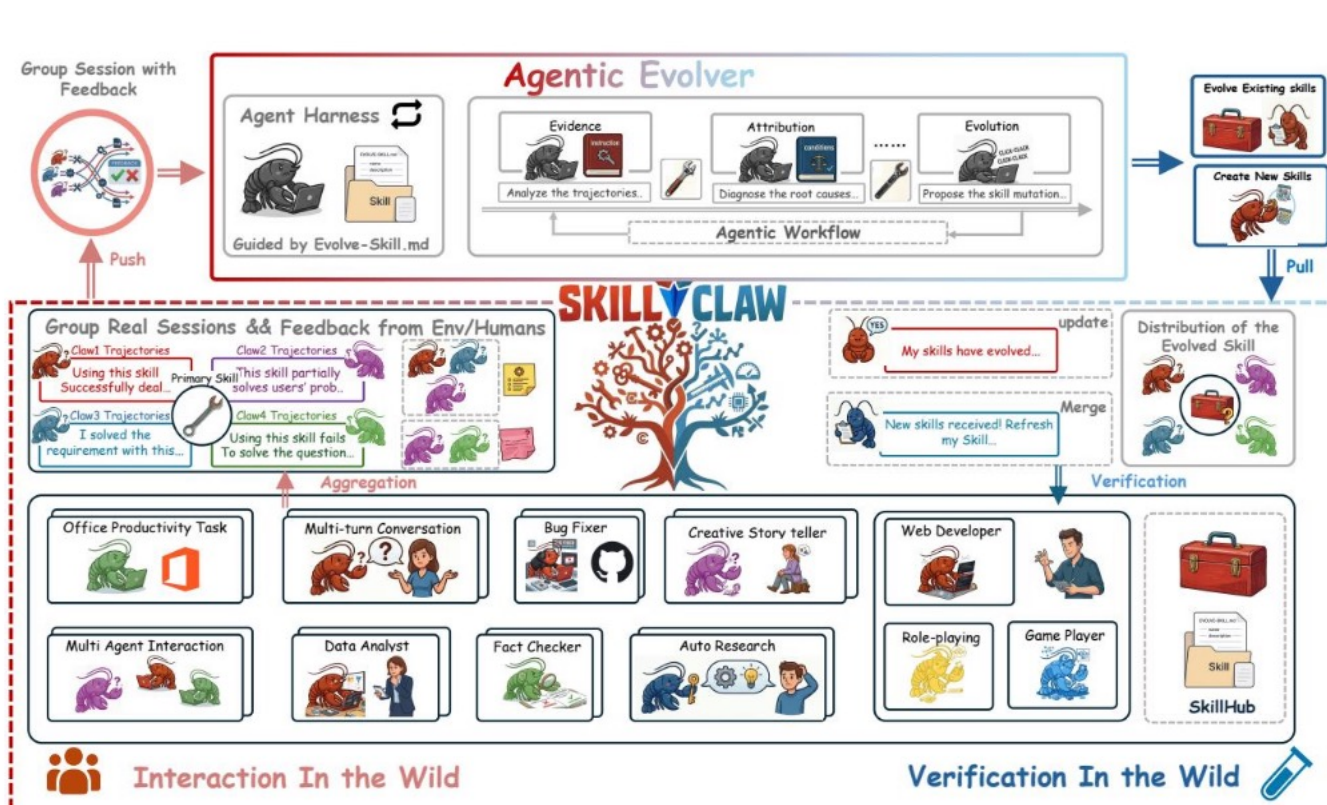
Users Rediscover Solutions Independently

The Status Quo

- Skills are manually installed from a centralized hub
- They remain completely static post-deployment
- Memory and skill libraries are tied to specific individuals

The Missing Link

- No system-level knowledge accumulation occurs
- The exact same failures are repeatedly observed across users
- Each user is forced to rediscover identical solutions independently



Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution

- **Evidence:** Analyze session trajectories to identify behavioral patterns
- **Attribution:** Diagnose root causes of failures and successes
- **Evolution:** Propose skill mutations (refining or creating skills)
- **Distribution:** Verified skills synchronize back to all agents

Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

Driven by open-ended agent reasoning over interaction evidence, not predefined update rules.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Different users exercising the exact same skill under diverse contexts produce highly complementary views of that skill's behavioral boundary, revealing both working conditions and breaking points.

Compatible Claw-style systems include: OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, and NemoClaw.

The Single User Constraint

"A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix."

AGGREGATION ENABLES TRUE GENERALIZATION

Case Study: Multimodal Creative Pipeline Skill

6 Days of Evolution Attempts

Conservative verification process: Only 1 of 6 candidates accepted.

Case Study: Git Push Auth-Fallback Skill

Iterative Refinement Over 6 Days

Successful iterative refinement: 3 of 6 days yielded accepted updates before plateauing.

Key Takeaways:

Three Properties That Distinguish SkillClaw

By shifting from static skill libraries to a continuously evolving ecosystem, we unlock powerful new capabilities for multi-user LLM architectures.

Implications for the Future

Agents that perpetually self-improve based on empirical evidence, eliminating the need for constant manual updates and curation.

01. Collective Evolution

Knowledge from individual interactions contributes directly to a shared, continuously improving skill ecosystem.

02. Fully Automatic

Skill evolution is driven entirely by runtime interaction evidence without manual curation or explicit user intervention.

03. Agentic Evolution Paradigm

Skill updates are produced through open-ended reasoning over evidence rather than via constrained, predefined update rules.